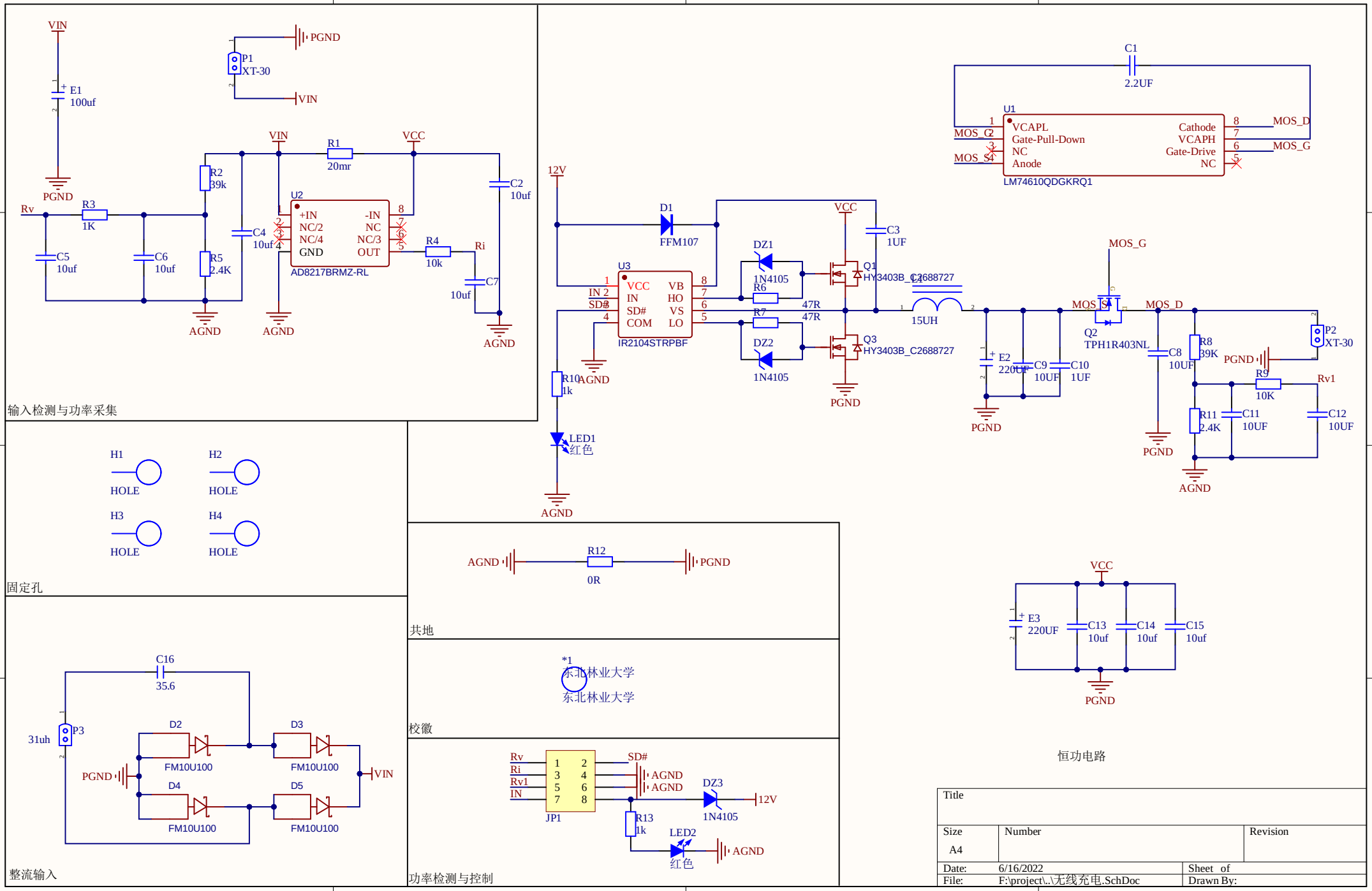
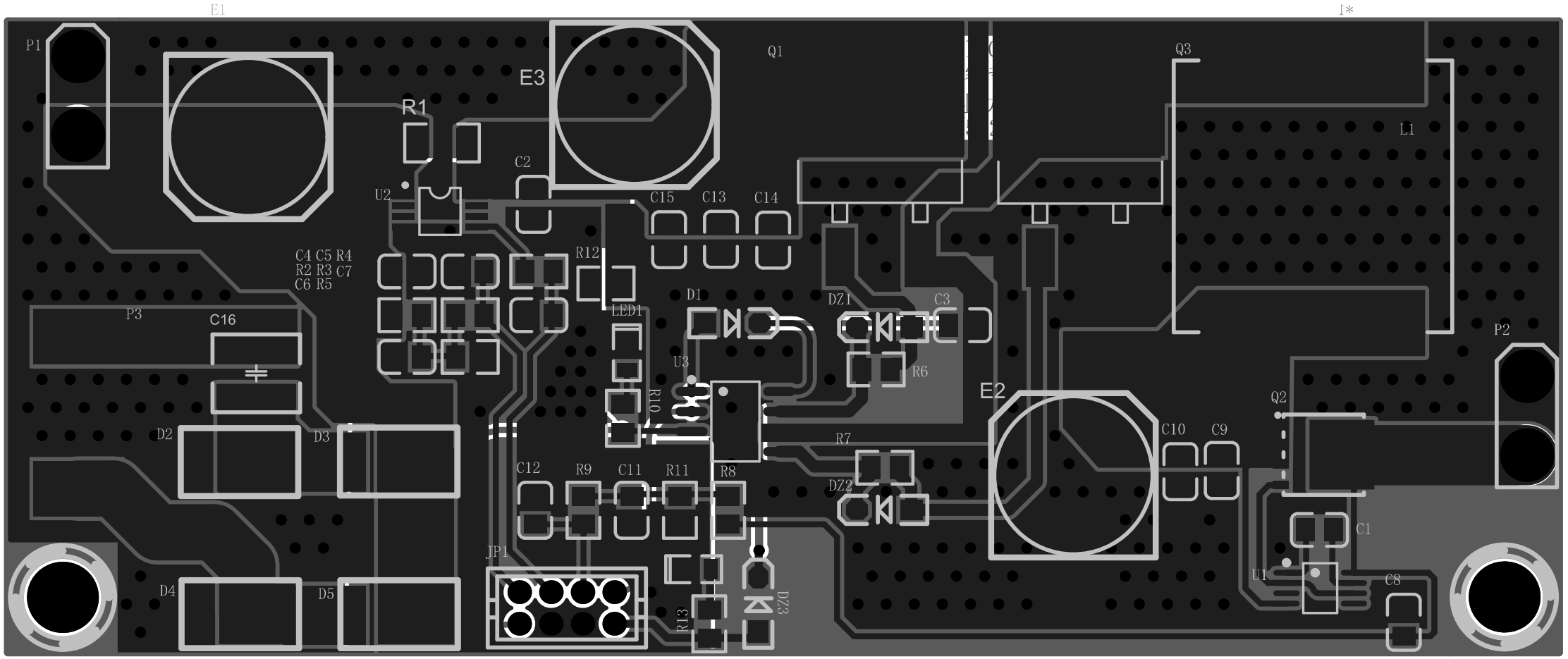

输入检测与功率采集
固定孔
共地
校徽
整流输入
功率检测与控制
恒功电路
VIN
PGND
P1
XT-30
E1
100uf
R1
20mr
VCC
R2
39k
R3
1K
Rv
C4
10uf
C5
10uf
C6
10uf
R5
2.4K
U2
+IN
NC/2
NC/4
GND
-IN
NC
NC/3
OUT
AD8217BRMZ-RL
R4
10k
Ri
C7
10uf
C2
10uf
AGND
12V
D1
FFM107
U3
VCC
IN
SD#
COM
VB
HO
VS
LO
IR2104STRPBF
R10
1k
LED1
红色
DZ1
1N4105
DZ2
1N4105
R6
47R
R7
47R
Q1
HY3403B_C2688727
Q3
HY3403B_C2688727
C3
1UF
L1
15UH
E2
220UF
C9
10UF
C10
1UF
C1
2.2UF
U1
VCAPL
Gate-Pull-Down
NC
Anode
Cathode
VCAPH
Gate-Drive
NC
MOS_G2
MOS_S4
MOS_D
MOS_G
LM74610QDGKRQ1
MOS_G
MOS_S
MOS_D
Q2
TPH1R403NL
C8
10UF
R8
39K
R9
10K
Rv1
R11
2.4K
C11
10UF
C12
10UF
P2
XT-30
H1
HOLE
H2
HOLE
H3
HOLE
H4
HOLE
R12
0R
*1
东北林业大学
东北林业大学
C16
35.6
P3
31uh
D2
FM10U100
D3
FM10U100
D4
FM10U100
D5
FM10U100
JP1
SD#
DZ3
1N4105
R13
1k
LED2
红色
E3
220UF
C13
10uf
C14
10uf
C15
10uf
Title
Size
A4
Number
Revision
Date:
6/16/2022
Sheet of
File:
F:\project\..\无线充电.SchDoc
Drawn By:


E1
I*
P1
E3
Q1
Q3
R1
L1
C2
U2
C15
C13
C14
C4 C5 R4
R2 R3 C7
C6 R5
R12
D1
DZ1
C3
P3
C16
LED1
U3
R6
P2
E2
R10
Q2
D2
D3
R7
C10
C9
C12
R9
C11
R11
R8
DZ2
C1
JP1
U1
C8
D4
D5
R13
DZ3

Comment	Description	Designator	Footprint	LibRef	Quantity
东北林业大学		*1	校徽300mil	东北林业大学	1
2.2UF	贴片电容	C1	0805_C	0805 10nF (103) 10% 50V	1
10uf	贴片电容	C2, C4, C5, C6, C7, C8, C9, C11, C12, C13, C14, C15	0805_C	0805 10nF (103) 10% 50V	12
1UF	贴片电容	C3, C10	0805_C	0805 10nF (103) 10% 50V	2
35.6	无极性贴片电容	C16	C 1210_L	C	1
FFM107	快恢复整流二极管	D1	SOD-123	FR107w	1
FM10U100		D2, D3, D4, D5	TO-277B_3P-L6.1-W4.3-LS7.0-TR	FM10U100	4
1N4105	11V/0.5W稳压管	DZ1, DZ2, DZ3	SOD-123	1N4105	3
100uf	贴片电解电容	E1	CM F(10*10.2)	C-CM	1
220UF	贴片电解电容	E2, E3	CM F(10*10.2)	C-CM	2
HOLE	固定孔	H1, H2, H3, H4	LQ_SCROW3&5MM	HOLE	4
Header 4X2	4*2P接插件	JP1	CONN-TH_A2006WV-2X4P	Header 4X2	1
15UH	一体化成型电感	L1	IND-SMD_L17.5-W17.2	L	1
红色	0805LED	LED1, LED2	LED 0805B	LED-5MM	2
XT-30	接口	P1, P2	xt-30 - duplicate	XT-30_1	2
31uh	接口	P3	线圈	线圈	1
HY3403B_C2688727		Q1, Q3	TO-263-2_L10.1-W9.1-P5.08-LS15.2-BR	HY3403B_C2688727	2
TPH1R403NL		Q2	SOP Advance	TPH1R403NL	1
20mr	贴片电阻	R1	1206_R	1206 0.1Ω (R100) 1%	1
39k	贴片电阻	R2, R8	0805_R	0805 10MΩ (1005) 1%	2
1K	贴片电阻	R3, R10, R13	0805_R	0805 10MΩ (1005) 1%	3
10k	贴片电阻	R4, R9	0805_R	0805 10MΩ (1005) 1%	2
2.4K	贴片电阻	R5, R11	0805_R	0805 10MΩ (1005) 1%	2
47R	贴片电阻	R6, R7	0805_R	0805 10MΩ (1005) 1%	2
0R	贴片电阻	R12	0805_R	0805 10MΩ (1005) 1%	1
LM74610QDGKRQ1		U1	VSSOP-8_L3.0-W3.0-P0.65-LS5.0-BL	LM74610QDGKRQ1	1
AD8217BRMZ-RL		U2	MSOP-8_L3.0-W3.0-P0.65-LS4.9-BL	AD8217BRMZ-RL	1
IR2104STRPBF		U3	SOIC-8_L4.9-W3.9-P1.27-LS6.0-BL	IR2104STRPBF	1